

# Atlas Embeddings: Exceptional Lie Groups from First Principles

## Mathematical Foundations and Main Theorems

UOR FOUNDATION

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We present a publication-ready specification of the  $E_8$  root system with a deterministic paste-stable ordering, an injective reconstructive labeling, and a category of *ResGraph* objects capturing induced substructures and symmetries. Simple roots are given in a Bourbaki-consistent realization (trivalent at node 4) with inner product  $\langle \alpha_i, \alpha_j \rangle = -1$  on Dynkin edges and 2 on the diagonal.

A  $\Sigma$ -calculus encodes constructions via three primitives: parabolic restriction  $\text{Par}_I$ , admissible fold  $\text{fold}_\Gamma$  (including quotient folds), and junction tagging  $\text{Aug}_J$ . Using these, we supply five paste-stable  $\Sigma$ -terms whose reflection-closures recover the classical types inside  $E_8$ :  $G_2$  from a  $D_4$  triality fold,  $F_4$  from an  $E_6$  quotient fold,  $E_6$  from a standard parabolic,  $E_7$  from a parabolic with admissible augmentation, and  $E_8$  from the Atlas simple system itself.

Each construction is backed by machine checks: Cartan recovery, determinant and rank, integrality of Cartan integers on the closure, Coxeter parameters, root counts, and Weyl orders. Verified outputs are:  $G_2$  (12 roots,  $|W| = 12$ ),  $F_4$  (48 roots,  $|W| = 1152$ ),  $E_6$  (72 roots,  $|W| = 51,840$ ),  $E_7$  (126 roots,  $|W| = 2,903,040$ ), and  $E_8$  (240 roots,  $|W| = 696,729,600$ ). The framework fixes conventions end-to-end, yields reproducible code paths, and isolates admissibility conditions for folds and augmentations. This provides a minimal, verifiable, and implementation-ready atlas for  $E_8$  and its induced subsystems.

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# 1 The Atlas of Resonance Classes

**Object.** A canonical 96–element set of resonance classes  $= \{v_1, \dots, v_{96}\} \subset \Lambda$  with a fixed adjacency predicate

$$(i \sim j) \iff v_i v_j =, \quad \in \{-1, +1\}.$$

The two sign choices are Weyl–equivariant; pick one and use it consistently.

**Multiplicity profile.** is split into two resonance levels with counts

$$32 \text{ elements at level 2,} \quad 64 \text{ elements at level 3.}$$

These labels are intrinsic to the atlas and do not rely on any particular coordinate chart.

**Embedding.** There exists an embedding  $\iota : \hookrightarrow \Lambda$  unique up to . Any two embeddings differ by an element of .

**Graph structure.** Let  $G()$  be the simple graph on 96 vertices with the above adjacency. Key invariants:

[nosep]

- Degree partition:  $\{d_a^{\times 32}, d_b^{\times 64}\}$  for some integers  $d_a \neq d_b$ .
- Induced exceptional subgraphs: counts  $N_{E_6}, N_{E_7}, N_{E_8}$  of  $E_6, E_7, E_8$  Dynkin diagrams.
- Automorphisms:  $\text{Aut}(G)$  contains the image of stabilizing .

**Verification sketch.** Given coordinates for in  $\Lambda$ :

[nosep]

1. Use exact rationals to compute the matrix  $M_{ij} = v_i v_j$ .
2. Form the adjacency matrix  $A$  by testing  $M_{ij} =$  for  $i \neq j$ .
3. Check the 96–way degree partition equals  $\{d_a^{\times 32}, d_b^{\times 64}\}$ .
4. Count induced  $E_6, E_7, E_8$  subgraphs as a sanity check of exceptional closures.
5. Record a certificate: the raw  $M$ , the edge set, and hashes of both.

**Definition 1** (Atlas Graph). The **Atlas**  $\mathcal{A} = (V_{\mathcal{A}}, E_{\mathcal{A}}, \lambda_{\mathcal{A}}, U_{\mathcal{A}})$  is a 96–vertex graph with:

- $V_{\mathcal{A}}$ : 96 vertices representing resonance classes
- $E_{\mathcal{A}} \subseteq V_{\mathcal{A}} \times V_{\mathcal{A}}$ : edges defined by the adjacency predicate

- $\lambda_A : V_A \rightarrow R_{E_8}$ : vertex labeling into  $E_8$  roots
- $U_A \subseteq V_A$ : distinguished unity vertices

**Definition 2** (Adjacency Predicate). For  $t \in \{0, \pm 1\}$ , define:

$$\Phi(t) \iff (t = -1)$$

Two vertices  $v, w \in V_A$  are adjacent iff  $\langle \lambda_A(v), \lambda_A(w) \rangle = -1$ .

## 1.1 The Category of Resonance Graphs

**Definition 3** (Resonance Graph Object). A **ResGraph object** is a tuple  $\mathcal{G} = (V, E, \lambda, U)$  where:

1.  $V$  is a finite set of vertices
2.  $E \subseteq \{\{v, w\} : v, w \in V, v \neq w\}$  is a set of edges
3.  $\lambda : V \rightarrow R_{E_8}$  labels vertices with  $E_8$  roots
4.  $U \subseteq V$  is a distinguished unity subset
5. Adjacency rule:  $\{v, w\} \in E$  iff  $\Phi(\langle \lambda(v), \lambda(w) \rangle)$

**Definition 4** (ResGraph Morphism). A morphism  $\varphi : (V, E, \lambda, U) \rightarrow (V', E', \lambda', U')$  is a function  $\varphi : V \rightarrow V'$  such that:

1. (Graph homomorphism)  $\{v, w\} \in E \Rightarrow \{\varphi(v), \varphi(w)\} \in E'$
2. (Label compatibility)  $\exists g \in W(E_8)$  such that  $\lambda'(\varphi(v)) = g \cdot \lambda(v)$  for all  $v \in V$
3. (Unity preservation)  $\varphi(U) \subseteq U'$

## 1.2 Main Theorem: Atlas to $E_8$ Embedding

**Theorem 5** (Atlas Embedding). There exists an injective map  $f : V_A \rightarrow R_{E_8}$  such that:

1. For all  $v, w \in V_A$ ,  $\{v, w\} \in E_A$  iff  $\langle f(v), f(w) \rangle = -1$
2.  $|\text{im}(f)| = 96$
3. (Uniqueness up to Weyl group) If  $f'$  also satisfies (1) and (2), then there exists  $w \in W(E_8)$  with  $f' = w \circ f$

*Proof Outline.* **Existence:** Construct  $f$  explicitly by providing coordinates for all 96 vertices in the canonical  $E_8$  basis. Verify adjacency by checking all  $\binom{96}{2}$  pairs satisfy  $\Phi$ .

**Uniqueness:**

1. Fix a seed vertex  $v_0$  and normalize  $f(v_0)$  to a chosen root via Weyl action

2. Neighbors of  $v_0$  must map to the 56 roots with inner product  $-1$  with  $f(v_0)$
3. Local adjacency patterns force unique assignments via connectedness of  $\mathcal{A}$
4. Any alternative embedding differs by a global Weyl transformation

□

### 1.3 Categorical Operations in ResGraph

**Definition 6** (Reflection Closure). For  $\mathcal{G} = (V, E, \lambda, U)$ , define  $\text{RC}(\mathcal{G})$  as the smallest ResGraph containing  $\mathcal{G}$  whose label set is closed under reflections:

$$\Lambda_{k+1} = \Lambda_k \cup \{s_\alpha(\beta) : \alpha, \beta \in \Lambda_k\} \cap R_{E_8}$$

with  $\Lambda_0 = \lambda(V)$ .

**Definition 7** (Resonance Product). For  $\mathcal{G}_1, \mathcal{G}_2$ , define the partial sum graph:

$$\mathcal{G}_1 \boxtimes \mathcal{G}_2 = \{(v_1, v_2) : \lambda_1(v_1) + \lambda_2(v_2) \in R_{E_8}\} / \sim$$

where  $(v_1, v_2) \sim (v'_1, v'_2)$  if their label sums coincide.

**Definition 8** (Quotient by Symmetry). For  $H \leq \text{Aut}(\mathcal{G})$ , define  $\mathcal{G}/H$  with vertices as  $H$ -orbits, labels by representatives, and edges between orbits when any representatives are adjacent.

### 1.4 Exceptional Group Constructions

**Theorem 9** ( $G_2$  Construction). The exceptional group  $G_2$  emerges as:

$$G_2 = \text{RC}(\mathcal{A} \boxtimes C_3) / H$$

where  $C_3$  is a 3-element resonance graph and  $H$  an appropriate symmetry group. The construction yields:

- 12 roots (6 short, 6 long)
- Rank 2, Cartan matrix  $\begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$
- Weyl group order 12

**Theorem 10** ( $F_4$  Construction).

$$F_4 = \text{RC}(\mathcal{A}/\tau)$$

where  $\tau$  is the mirror symmetry involution, yielding:

- 48 roots (24 short, 24 long)
- Rank 4

- Weyl group order 1152

**Theorem 11** ( $E_6, E_7, E_8$  Constructions). •  $E_6 = \text{RC}(\mathcal{A}[\ell = c])$  via hyperplane filtration

- $E_7 = \text{Aug}(\mathcal{A})$  via root addition closure
- $E_8 = \text{RC}(\mathcal{A})$  via full reflection closure

## 1.5 Initiality and Universal Property

**Conjecture 12** (Atlas Initiality). *The Atlas  $\mathcal{A}$  is initial in **ResGraph**: for every object  $\mathcal{G} \in \text{ResGraph}$ , there exists a unique morphism  $\varphi : \mathcal{A} \rightarrow \mathcal{G}$ .*

**Definition 13** (Precursor Category). Define **ResGraph**<sub>0</sub> with objects  $(V, E, U)$  (forgetting labels) and morphisms preserving graph structure and unity. There is a faithful functor  $J : \text{ResGraph} \rightarrow \text{ResGraph}_0$ .

**Lemma 14** (Lifting Lemma). *Any **ResGraph**<sub>0</sub>-morphism  $f : J(\mathcal{A}) \rightarrow J(\mathcal{G})$  lifts uniquely to a **ResGraph**-morphism  $f : \mathcal{A} \rightarrow \mathcal{G}$  if  $\mathcal{G}$  satisfies the  $E_8$ -liftability condition.*

## 1.6 Action Functional and Variational Foundation

**Definition 15** (Flag-Resonance Complex). For  $\mathcal{G} = (V, E, \lambda, U)$ , define the cellular complex  $\text{FR}(\mathcal{G})$  by:

- 0-cells:  $V$
- 1-cells:  $E$
- $k$ -cells:  $\text{clique}_{k+1}(\mathcal{G})$  (cliques of size  $k+1$ )

**Definition 16** (Action Functional). For  $\varphi \in C^{k-1}(\text{FR}(\mathcal{G}); A)$ , define:

$$S_k[\varphi] = \sum_{\sigma \in \text{FR}^{(k)}} \varphi(\partial\sigma)$$

*The Atlas emerges as stationary configuration of  $S_k$  for appropriate  $k$  and coefficient group  $A$ .*

## 1.7 Computational Verification Framework

**Theorem 17** (Exact Arithmetic Verification). *All constructions and verifications use exact rational arithmetic:*

1. Vertex coordinates in  $\mathbb{Q}$  or  $\frac{1}{2}\mathbb{Z}$
2. Inner products computed exactly via  $\langle x, y \rangle = \sum x_i y_i$
3. Adjacency verified by  $\Phi(\langle \lambda(v), \lambda(w) \rangle)$

#### 4. Cartan matrices derived from simple systems

**Proposition 18** (Reproducible Artifacts). *The theory is computationally certified by:*

- Complete vertex/edge listings for  $\mathcal{A}$
- Explicit coordinate tables for all exceptional groups
- Verification scripts confirming all mathematical claims
- Flag complex cell counts justifying the 12,288-cell structure

## 2 The $E_8$ Root System

The classification of simple Lie algebras reveals five exceptional cases:  $G_2$ ,  $F_4$ ,  $E_6$ ,  $E_7$ , and  $E_8$ . While these have been extensively studied since their discovery by Killing and Cartan, their emergence from first principles remains mysterious. This work presents a unified construction from a single combinatorial object: the

**Definition 19** ( $E_8$  Root System). *The  $E_8$  root system  $R_{E_8} \subset \mathbb{R}^8$  consists of 240 vectors of squared norm 2:*

- 112 integer roots:  $\pm e_i \pm e_j$  for  $1 \leq i < j \leq 8$
- 128 half-integer roots:  $(\pm \frac{1}{2}, \pm \frac{1}{2}, \dots, \pm \frac{1}{2})$  with an even number of minus signs

*The inner product between distinct roots takes values in  $\{0, \pm 1\}$ .*

**Definition 20** (Weyl Group). *The Weyl group  $W(E_8)$  is generated by reflections  $s_\alpha(x) = x - \frac{2\langle x, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha$  for  $\alpha \in R_{E_8}$ .*

### 2.1 Root System $G_2$ (product)

**$\Sigma$ -term.** Work in Bourbaki  $E_8$  with simple roots  $\{\alpha_i\}_{i=1}^8$  and  $\langle \alpha_i, \alpha_j \rangle = -1$  exactly along Dynkin edges. Inside  $E_8$ , the subset  $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$  is a  $D_4$  with hub  $\alpha_3$  and legs  $\alpha_1, \alpha_2, \alpha_4$ . Define the  $\Sigma$ -term

$$t_{G_2} := \text{fold}_{C_3}(\text{Par}_{\{1,2,4\}}; \text{Aug}_3)(\text{Atlas marking}),$$

so that

$$\beta_\ell := \alpha_1 + \alpha_2 + \alpha_4, \quad \beta_s := \alpha_3.$$

**Cartan check.** With  $\langle \alpha_i, \alpha_i \rangle = 2$ , legs pairwise orthogonal, and each leg meeting the hub once,

$$\|\beta_\ell\|^2 = 6, \quad \|\beta_s\|^2 = 2, \quad \langle \beta_\ell, \beta_s \rangle = -3,$$

hence

$$A = \begin{pmatrix} 2 & \frac{2\langle \beta_\ell, \beta_s \rangle}{\|\beta_s\|^2} \\ \frac{2\langle \beta_s, \beta_\ell \rangle}{\|\beta_\ell\|^2} & 2 \end{pmatrix} = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix},$$

the crystallographic Cartan matrix of type  $G_2$ .



**Angle and Coxeter parameter.**

$$\cos \theta = \frac{\langle \beta_s, \beta_\ell \rangle}{\|\beta_s\| \|\beta_\ell\|} = -\frac{3}{\sqrt{12}} = -\frac{\sqrt{3}}{2} \Rightarrow \theta = 150^\circ.$$

For simple roots,  $\theta = \pi - \frac{\pi}{m}$ , so  $m = \frac{\pi}{\pi-\theta} = 6$ .

**Root strings.** With  $\alpha = \beta_s$  and  $\beta = \beta_\ell$ :

$$\langle \beta, \alpha^\vee \rangle = -3 \Rightarrow \alpha\text{-string through } \beta \text{ has length } 1 - (-3) = 4,$$

$$\langle \alpha, \beta^\vee \rangle = -1 \Rightarrow \beta\text{-string through } \alpha \text{ has length } 1 - (-1) = 2,$$

matching  $G_2$  string lengths and ensuring crystallographic closure.

**Proposition 21.** *The reflection-closure of  $\{\beta_\ell, \beta_s\}$  in the  $E_8$  ambient is the  $G_2$  root system. The Weyl group is dihedral of order 12.*

*Proof outline.* Triality  $C_3$  permutes the three  $D_4$  legs, and folding along  $C_3$  is admissible in simply-laced type because nodes in the same orbit are nonadjacent. The Gram data above yield the  $G_2$  Cartan matrix. The angle gives  $m = 6$ , so the rank-2 Coxeter group generated by  $s_{\beta_s}, s_{\beta_\ell}$  is  $D_{12}$  of order 12. Root-string lengths  $(4, 2)$  confirm crystallographic closure with long:short norm ratio  $3 : 1$ .  $\square$

## 2.2 Root System $F_4$ (quotient fold of an $E_6 \subset E_8$ slice)

**$\Sigma$ -term.** Let  $\{\alpha_i\}_{i=1}^8$  be the Bourbaki  $E_8$  simple roots with  $\langle \alpha_i, \alpha_j \rangle = -1$  on Dynkin edges. Inside  $E_8$ , take a symmetric  $E_6$  slice written as a chain with a central node and an attached arm:

$$L - \ell - \boxed{c} - r - R, \quad u \text{ attached to } c.$$

Let  $\Gamma = C_2$  act by  $L \leftrightarrow R$  and  $\ell \leftrightarrow r$ , fixing  $c, u$ . Define the quotient (orbit-average) fold

$$t_{F_4} := \text{fold}_{C_2}^{\text{quot}}(\text{Par}_{E_6}; \text{Aug}_{\{\text{center}, \text{arm}\}})(\text{Atlas}), \quad p(O) = \frac{1}{|O|} \sum_{\alpha \in O} \alpha.$$

**Folded generators.**

$$\beta_1 = \frac{1}{2}(L + R), \quad \beta_2 = \frac{1}{2}(\ell + r), \quad \beta_3 = c, \quad \beta_4 = u.$$

**Gram data in  $E_8$ .** Lengths:  $\|\beta_1\|^2 = \|\beta_2\|^2 = 1$  (short),  $\|\beta_3\|^2 = \|\beta_4\|^2 = 2$  (long). Pairings:

$$\langle \beta_1, \beta_2 \rangle = -\frac{1}{2}, \quad \langle \beta_2, \beta_3 \rangle = -1, \quad \langle \beta_3, \beta_4 \rangle = -1,$$

others 0.

**Cartan matrix.** With  $A_{ij} = 2\langle\beta_i, \beta_j\rangle/\|\beta_j\|^2$ ,

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -2 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix},$$

which is the  $F_4$  Cartan (double bond between nodes 2 and 3, arrow toward the short root).

**Angles and strings.**  $m_{12} = 3$  (short–short,  $\cos 120^\circ$ ),  $m_{23} = 4$  (short–long),  $m_{34} = 3$  (long–long). Root strings across  $(\beta_2, \beta_3)$  have lengths 2 and 3 since  $\langle\beta_2, \beta_3^\vee\rangle = -1$  and  $\langle\beta_3, \beta_2^\vee\rangle = -2$ .

**Proposition 22.** *The reflection-closure of  $\{\beta_1, \beta_2, \beta_3, \beta_4\}$  is the  $F_4$  root system. The Weyl group has order 1152.*

*Proof outline.* The  $C_2$  involution swaps two symmetric arms of the  $E_6$  slice, so the quotient fold via orbit-averaging is admissible. The Gram data yield the  $F_4$  Cartan above. The adjacent Coxeter parameters  $(3, 4, 3)$  match  $F_4$ . Root-string lengths  $(2, 3)$  across the double bond certify crystallographic closure with the expected short/long structure. Therefore the reflection-closure is  $F_4$  and the Weyl group order is 1152.  $\square$

## 2.3 Root System $E_6$ (filtration)

**$\Sigma$ -term.** Work in Bourbaki  $E_8$  with  $\langle\alpha_i, \alpha_j\rangle = -1$  on Dynkin edges. Select the standard  $E_6$  parabolic and tag the central node for checks:

$$t_{E_6} := \text{Par}_I \circ \text{Aug}_{\{4\}} \text{ (Atlas)}, \quad I = \{1, 2, 3, 4, 5, 6\}.$$

As a filtration one may use

$$E_8 \xrightarrow{\text{Par}_{\{1, \dots, 7\}}} E_7 \xrightarrow{\text{Par}_{\{1, 2, 3, 4, 5, 6\}}} E_6.$$

**Simple system produced.** Set  $\beta_i := \alpha_i$  for  $i \in I$ . Adjacencies (trivalent node at 4):

$$1-3, \quad 3-4, \quad 4-5, \quad 5-6, \quad 2-4,$$

others 0. All  $\|\beta_i\|^2 = 2$ .

**Cartan matrix.** With  $A_{ij} = 2\langle\beta_i, \beta_j\rangle/\|\beta_j\|^2$ ,

$$A = \begin{pmatrix} 2 & 0 & -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 & 0 & 0 \\ -1 & 0 & 2 & -1 & 0 & 0 \\ 0 & -1 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}, \quad \det A = 3, \text{ rank} = 6.$$

**Reflection-closure and size.** Since  $A$  is simply-laced, crystallographic, and indecomposable, the reflection-closure is the  $E_6$  root system with  $|\Phi| = 72$  and Coxeter number  $h = 12$ .

**Weyl order.** Exponents  $(1, 4, 5, 7, 8, 11)$  give

$$|W(E_6)| = \prod_{i=1}^6 (m_i + 1) = 2 \cdot 5 \cdot 6 \cdot 8 \cdot 9 \cdot 12 = 51,840.$$

## 2.4 Root System $E_7$ (augmentation)

**$\Sigma$ -term.** Work in Bourbaki  $E_8$  with  $\langle \alpha_i, \alpha_j \rangle = -1$  on Dynkin edges. Select the standard  $E_7$  parabolic and tag the junction (node 4) for checks:

$$t_{E_7} := \text{Aug}_{\{4\}} \circ \text{Par}_I (\text{Atlas}), \quad I = \{1, 2, 3, 4, 5, 6, 7\}.$$

The augmentation is admissible: it adds no edges and preserves integrality.

**Simple system produced.** Set  $\beta_i := \alpha_i$  for  $i \in I$ . Adjacencies (trivalent at 4):

$$1-3, \quad 3-4, \quad 4-5, \quad 5-6, \quad 6-7, \quad 2-4,$$

others 0. All  $\|\beta_i\|^2 = 2$ .

**Cartan matrix.** With  $A_{ij} = 2\langle \beta_i, \beta_j \rangle / \|\beta_j\|^2$ ,

$$A = \begin{pmatrix} 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}, \quad \det A = 2, \text{ rank} = 7.$$

**Reflection-closure and size.** Since  $A$  is simply-laced, crystallographic, indecomposable, the reflection-closure is the  $E_7$  root system with  $|\Phi| = 126$  and Coxeter number  $h = 18$ .

**Weyl order.** Exponents  $(1, 5, 7, 9, 11, 13, 17)$  give

$$|W(E_7)| = \prod_{i=1}^7 (m_i + 1) = 2 \cdot 6 \cdot 8 \cdot 10 \cdot 12 \cdot 14 \cdot 18 = 2,903,040.$$

Work in  $\mathbb{R}^8$  with the standard dot product  $\langle x, y \rangle = \sum_{i=1}^8 x_i y_i$ . Define the even unimodular lattice

$$E_8 = \{x \in \mathbb{Z}^8\} \cup \left\{x \in \left(\mathbb{Z} + \frac{1}{2}\right)^8 : \sum_{i=1}^8 x_i \in 2\mathbb{Z}\right\}.$$

The root set is

$$R = \{\alpha \in E_8 : \langle \alpha, \alpha \rangle = 2\}, \quad |R| = 240.$$

Two disjoint types:

Type I: all permutations of  $(\pm 1, \pm 1, 0, \dots, 0)$  (no sign-parity restriction)  $\Rightarrow$  112 roots,

Type II:  $(\pm \frac{1}{2}, \dots, \pm \frac{1}{2})$  with even parity (even # of  $-1$  entries)  $\Rightarrow$  128 roots.

Possible inner products among roots are  $\{-2, -1, 0, 1, 2\}$ .

## 2.5 Cartan Matrix, Simple Roots, and Weyl Group

Adopt the Bourbaki  $E_8$  Cartan matrix with the trivalent node at 3:

$$A = \begin{pmatrix} 2 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

A compatible simple system  $\{\alpha_i\}_{i=1}^8 \subset \mathbb{R}^8$  is:

$$\begin{aligned} \alpha_1 &= \frac{1}{2}(1, -1, -1, -1, -1, -1, -1, 1), & \alpha_2 &= (1, 1, 0, 0, 0, 0, 0, 0), \\ \alpha_3 &= (-1, 0, 1, 0, 0, 0, 0, 0), & \alpha_4 &= (0, 0, -1, 1, 0, 0, 0, 0), \\ \alpha_5 &= (0, 0, 0, -1, 1, 0, 0, 0), & \alpha_6 &= (0, 0, 0, 0, -1, 1, 0, 0), \\ \alpha_7 &= (0, 0, 0, 0, 0, -1, 1, 0), & \alpha_8 &= (0, 0, 0, 0, 0, 0, -1, 1). \end{aligned}$$

Then  $\langle \alpha_i, \alpha_i \rangle = 2$  and  $\langle \alpha_i, \alpha_j \rangle = -1$  exactly on the edges of  $A$ ; all other off-diagonals are 0. Define reflections

$$s_i(x) = x - \frac{2\langle x, \alpha_i \rangle}{\langle \alpha_i, \alpha_i \rangle} \alpha_i \quad (\langle \alpha_i, \alpha_i \rangle = 2),$$

and the Weyl group  $W(E_8) = \langle s_1, \dots, s_8 \rangle$ , which acts transitively on  $R$ .

## 2.6 Paste-Stable Ordering $\iota$

We specify a deterministic total order  $\iota : R \rightarrow \{1, \dots, 240\}$ :

[itemsep=2pt,topsep=4pt,leftmargin=16pt]

1. **Type I ordering.** Write  $\alpha = \epsilon_i e_i + \epsilon_j e_j$  with  $i < j$  and  $\epsilon_i, \epsilon_j \in \{\pm 1\}$ . Order by the tuple

$$(i, j, \epsilon_i, \epsilon_j) \quad \text{under the bit order } -1 < +1.$$

2. **Type II ordering.** Use the sign vector  $\sigma(\alpha) \in \{\pm 1\}^8$  (even parity). Order lexicographically on  $(\sigma_1, \dots, \sigma_8)$  with  $-1 < +1$ .

3. Do not identify  $\alpha$  with  $-\alpha$  for ordering.

## 2.7 Adjacency Predicate and Root Graph

Define the  $W(E_8)$ -invariant adjacency

$$\Phi(\alpha, \beta) \iff \langle \alpha, \beta \rangle = -1.$$

This yields a simple undirected 56-regular graph on vertex set  $R$ .

## 2.8 Injective, Reconstructive Labels

Define  $\lambda : R \rightarrow \{I, II\} \times \mathcal{D}$  by

$$\lambda(\alpha) = \begin{cases} (I, (i, j), (\epsilon_i, \epsilon_j)) & \text{if } \alpha = \epsilon_i e_i + \epsilon_j e_j, \ i < j, \\ (II, \sigma(\alpha)) & \text{if } \alpha \in \{\pm \frac{1}{2}\}^8 \text{ with even parity,} \end{cases}$$

where  $\sigma(\alpha)$  is the 8-bit sign vector. The reconstruction map

$$\text{rec} \circ \lambda = \text{id}_V \quad \text{for any } V \subseteq R$$

is immediate from the data.

## 2.9 ResGraph Objects, Morphisms, and Atlas

### Objects

A *ResGraph* is a tuple  $G = (V, E, \lambda, U)$  with

$$\begin{aligned} V &\subseteq R \text{ specified by indices } I \subseteq \{1, \dots, 240\} : & V &= \{\alpha \in R : \iota(\alpha) \in I\}, \\ E &= \{\{\alpha, \beta\} \subseteq V : \Phi(\alpha, \beta)\}, & \lambda &= \text{labels above, restricted to } V, \\ U &= \{w|_V : w \in W(E_8), w(V) = V\} \subseteq \text{Aut}(G). \end{aligned}$$

### Morphisms

A morphism  $f : (V, E, \lambda, U) \rightarrow (V', E', \lambda', U')$  satisfies:

[itemsep=2pt,topsep=4pt,leftmargin=16pt]

1. **Adjacency reflection:**  $\Phi(\alpha, \beta) \iff \Phi'(f(\alpha), f(\beta)).$
2. **Label preservation:**  $\lambda' \circ f = \lambda.$
3. **Equivariance:**  $\forall u \in U \exists u' \in U' \text{ s.t. } f \circ u = u' \circ f.$

Items (1) and (2) imply  $f$  is injective on vertices.

## Atlas and Sub-Atlas

[itemsep=2pt,topsep=2pt,leftmargin=16pt]

- **Canonical full instance:**  $V = R$ ,  $E = \{\{\alpha, \beta\} : \langle \alpha, \beta \rangle = -1\}$ , reconstructive  $\lambda$ , and  $U = W(E_8)$ .
- **Induced sub-Atlas on**  $I \subseteq \{1, \dots, 240\}$ : restrict  $V, E, \lambda$ ; set  $U = \{w|_V : w \in W(E_8), w(V) = V\}$ .

## 2.10 Example: The $E_7$ Subsystem

Let

$$V = \{\alpha \in R : \langle \alpha, \alpha_8 \rangle = 0\} \quad (\text{roots orthogonal to } \alpha_8),$$

and  $E = \{\{\alpha, \beta\} \subset V : \langle \alpha, \beta \rangle = -1\}$ , with  $\lambda$  restricted to  $V$  and

$$U = \{w|_V : w \in W(E_8), w(V) = V\} \cong W(E_7).$$

## 2.11 Automorphisms

The  $E_8$  Dynkin diagram has trivial diagram automorphism. The longest Weyl group element  $w_0$  acts as  $-\text{Id}$  on  $\mathbb{R}^8$ , so  $-\text{Id} \in W(E_8)$ . Hence

$$\text{Aut}(R, \Phi) = W(E_8).$$

## 2.12 Validation Checklist

For implementation, verify:

[itemsep=2pt,topsep=4pt,leftmargin=16pt]

1. **Cartan verification:**  $G_{ij} = \langle \alpha_i, \alpha_j \rangle$  equals  $A$ .
2. **Root enumeration:** Enumerate  $R$  from  $E_8$ ; check  $|R| = 240$  and  $\langle \alpha, \alpha \rangle = 2$  for all  $\alpha \in R$ .
3. **Adjacency:** Build  $\Phi$ ; every vertex has degree 56.
4. **Weyl-invariance and transitivity:**  $\Phi(\alpha, \beta) = \Phi(w\alpha, w\beta)$  for  $w \in W(E_8)$  and  $W(E_8)$  acts transitively on  $R$ .
5. **Label reconstruction:** For  $V \subseteq R$ , confirm  $\text{rec} \circ \lambda = \text{id}_V$ .

### 3 Group Action, Orbit Decomposition, and Invariant Functionals

#### 3.1 Ambient Set and CRT Maps

Let  $P = \mathbb{Z}/48\mathbb{Z}$ ,  $B = \mathbb{Z}/256\mathbb{Z}$ , and  $G = P \times B$  so  $|G| = 48 \cdot 256 = 12,288$ . Write the Chinese Remainder isomorphism for  $P$  as

$$\psi : P \rightarrow \mathbb{Z}/16\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}, \quad \psi(p) = (a, c) = (p \bmod 16, p \bmod 3),$$

with inverse (encoding)

$$\varphi(a, c) \equiv 33a + 16c \pmod{48},$$

using  $3^{-1} \equiv 11 \pmod{16}$  and  $16^{-1} \equiv 1 \pmod{3}$ .

Represent  $a \in \mathbb{Z}/16\mathbb{Z}$  in binary as  $a = a_0 + 2a_1 + 4a_2 + 8a_3$  with  $a_i \in \{0, 1\}$ .

#### 3.2 Reference Subgroup and Action

Let  $V \cong (\mathbb{Z}/2\mathbb{Z})^3$  act by flipping bits  $a_0, a_1, a_2$  (bit  $a_3$  fixed). Let  $W \cong (\mathbb{Z}/2\mathbb{Z})^8$  act on bytes by XOR. Define

$$U_{\text{ref}} = V \times W \cong (\mathbb{Z}/2\mathbb{Z})^{11}.$$

For  $u = (v, w) \in U_{\text{ref}}$  and  $g = (p, b) \in G$ :

1.  $(a, c) = \psi(p)$ .
2.  $a' = (a \oplus v)$  on bits  $a_0, a_1, a_2$  with  $a_3$  unchanged.
3.  $b' = b \oplus w$ .
4.  $p' = \varphi(a', c)$  and  $u \cdot g = (p', b')$ .

The 11 standard generators are commuting, each of order 2, giving a simple undirected Cayley graph on  $G$ .

#### 3.3 Orbit Invariants, Freeness, and Tiling

Define the orbit invariant

$$\mathbf{o}(g) = (c, a_3) \in \mathbb{Z}/3\mathbb{Z} \times \{0, 1\}, \quad c = p \bmod 3, \quad a_3 = \lfloor a/8 \rfloor.$$

**Proposition 23.** *The action of  $U_{\text{ref}}$  on  $G$  is free. The invariant  $\mathbf{o}$  is preserved by  $U_{\text{ref}}$ . Each orbit has size  $2^{11} = 2048$  and is determined by  $(c, a_3)$ . There are exactly six orbits, with representatives*

$$S = \{(\varphi(0, c), 0), (\varphi(8, c), 0) : c \in \{0, 1, 2\}\}.$$

*Moreover, edges never cross orbits; each connected component of the Cayley graph equals a single orbit.*

*Sketch.* Nonzero  $v$  changes  $a$ ; nonzero  $w$  changes  $b$ ; hence stabilizers are trivial. The bits  $a_0, a_1, a_2$  and the byte are toggled freely, while  $c$  and  $a_3$  are fixed, so orbits have size  $2^{3+8} = 2048$  and are indexed by  $(c, a_3) \in \{0, 1, 2\} \times \{0, 1\}$ . The six anchors listed are in distinct orbits and tile  $G$ .  $\square$

### 3.4 Cayley Graph and Flag Complex

Let  $\Gamma$  be the Cayley graph of  $G$  with respect to the 11 generators. On each orbit,  $\Gamma$  is the 11-dimensional hypercube  $Q^{11}$ : per orbit  $n = 2^{11} = 2048$  vertices and  $m = 11 \cdot 2^{10} = 11264$  edges. The entire graph has six connected components, totaling

$$|V| = 12,288, \quad |E| = 67,584.$$

Hypercubes are triangle-free, thus the flag complex equals the 1-skeleton. For a graph,  $H_1$  is torsion-free and

$$\beta_1 = m - n + \#\text{components}.$$

Hence per orbit  $\beta_1 = 11264 - 2048 + 1 = 9217$ , and in total  $\beta_1 = 6 \cdot 9217 = 55,302$ . All higher Betti numbers vanish.

### 3.5 Schedule, Base Functional, and Invariance Scope

Let  $N = 768$  and the canonical fair schedule

$$\sigma(t) = (t \bmod 48, t \bmod 256), \quad t = 0, \dots, N-1.$$

For an affine observable  $f(p, b) = d + \alpha_p + \gamma_b$ , define the base functional

$$S_0(f) = \sum_{t=0}^{N-1} f(\sigma(t)) = 768 d + 16 \sum_{p \in P} \alpha_p + 3 \sum_{b \in B} \gamma_b.$$

**Proposition 24.**  *$S_0$  is invariant under arbitrary permutations of pages and bytes and under the  $U_{\text{ref}}$  action.*

[Limits for  $k \geq 1$ ] No polynomial weighting in  $t$  alone equalizes arithmetic-progression sums induced by  $\sigma$ . In particular, for  $\tau = t - (N-1)/2$ , the partial sums  $\sum_{t \equiv r \pmod{48}} \tau$  and  $\sum_{t \equiv s \pmod{256}} \tau$  are residue-dependent, so  $k \geq 1$  moment functionals are not invariant in general.

### 3.6 Permutation-Invariant Alternatives for First Moments

Define integer-scaled per-residue weights

$$w_p(t) = 2 \left\lfloor \frac{t}{48} \right\rfloor - 15, \quad w_b(t) = 2 \left\lfloor \frac{t}{256} \right\rfloor - 2,$$

so that for every page residue  $r$  and byte residue  $s$ ,

$$\sum_{i=0}^{15} w_p(r + 48i) = 0, \quad \sum_{i=0}^2 w_b(s + 256i) = 0.$$

Then the linear functionals

$$S_1^{\text{page}}(f) = \frac{1}{2} \sum_{t=0}^{N-1} w_p(t) f(\sigma(t)), \quad S_1^{\text{byte}}(f) = \frac{1}{2} \sum_{t=0}^{N-1} w_b(t) f(\sigma(t))$$

are invariant under arbitrary page permutations and byte permutations, respectively, and under the corresponding  $U_{\text{ref}}$  actions on  $P$  and  $B$ .



### 3.7 R96 Labeling

Define  $\lambda : B \rightarrow \{0, \dots, 95\}$  by  $\lambda(b) = b \bmod 96$ . Multiplicities are  $|\lambda^{-1}(r)| = 3$  for  $r \in \{0, \dots, 63\}$  and  $|\lambda^{-1}(r)| = 2$  for  $r \in \{64, \dots, 95\}$ , so  $64 \cdot 3 + 32 \cdot 2 = 256$ . Extend to  $G$  by composition with the byte projection; no equivariance beyond Weyl-invariance is asserted.

### 3.8 Operational Notes

Record the orbit tag  $\mathbf{o}(g)$  for indexing and caching. Since  $\mathbf{o}$  is invariant, edges never cross orbits and each component can be processed independently. Sparse  $\mathbb{F}_2$  or  $\mathbb{Z}$  homology on the 67,584-edge graph is straightforward; for graphs  $\beta_1$  is identical over  $\mathbb{Z}$  and  $\mathbb{F}_2$ .

### 3.9 E8-Based 96-Vertex Resonance Graph

Let  $R \subset \mathbb{R}^8$  be the  $E_8$  root system normalized by  $\langle \alpha, \alpha \rangle = 2$  for all  $\alpha \in R$ , so pairwise inner products lie in  $\{-2, -1, 0, 1, 2\}$ . Fix an injective indexing map

$$\iota : \{0, \dots, 95\} \longrightarrow R, \quad R_{96} := \text{im}(\iota),$$

and define the induced simple graph  $\Gamma_{E_8}^{(96)}$  by

$$V(\Gamma_{E_8}^{(96)}) = R_{96}, \quad \{\alpha, \beta\} \in E(\Gamma_{E_8}^{(96)}) \iff (\alpha \neq \beta \text{ and } \langle \alpha, \beta \rangle = +1).$$

No edges are added when  $\langle \alpha, \beta \rangle \in \{0, 1, 2, -2\}$ . This is the subgraph induced from the standard  $E_8$  root graph under the +1 adjacency rule.

**Adjacency conventions.** The full  $E_8$  ResGraph elsewhere uses  $-1$  adjacency (edges for  $\langle \alpha, \beta \rangle = -1$ ). The 96-vertex graph here intentionally uses  $+1$  adjacency. Both conventions are Weyl-equivariant. Note also that in this subsection all roots satisfy  $\langle \alpha, \alpha \rangle = 2$  by normalization.

**Integration with the R96 labeling.** Use the existing byte partition to index  $R_{96}$  via

$$\lambda_{E_8} : B \rightarrow R_{96}, \quad \lambda_{E_8}(b) := \iota(b \bmod 96).$$

Then  $|\lambda_{E_8}^{-1}(\alpha)| = 3$  for  $\alpha \in \iota(\{0, \dots, 63\})$  and  $|\lambda_{E_8}^{-1}(\alpha)| = 2$  for  $\alpha \in \iota(\{64, \dots, 95\})$ , matching the multiplicities  $64 \cdot 3 + 32 \cdot 2 = 256$ . No compatibility with the  $U_{\text{ref}}$  action is required or claimed.

**Independence from the Cayley layer.**  $\Gamma_{E_8}^{(96)}$  lives solely on the label set and does not modify the Cayley graph  $\Gamma$  or its flag complex. It is an auxiliary resonance structure replacing any “96-dimensional” phrasing with the precise term “96-vertex subgraph of the  $E_8$  root graph (adjacency  $\langle \alpha, \beta \rangle = +1$ )”.

**Optional normalization.** One may choose  $R_{96}$  closed under negation ( $\alpha \in R_{96} \Rightarrow -\alpha \in R_{96}$ ); this adds no edges because  $\langle \alpha, -\alpha \rangle = -2$ , but can be convenient for symmetry book-keeping. No further properties of  $R_{96}$  are assumed.

## 4 Atlas $\rightarrow E_8$ embedding: construction, invariants, uniqueness, and exact tests

### Construction

Work in the type-II root subsystem of the  $E_8$  lattice:

$$x = \frac{1}{2}(\pm 1, \dots, \pm 1) \in \mathbb{R}^8 \quad \text{with an even number of minus signs.}$$

Fix columns  $h_1, \dots, h_7 = e_i$ ,  $h_8 = \mathbf{1} \in \mathbb{F}_2^7$  with  $\sum_i h_i = 0$ . For  $u \in \mathbb{F}_2^7$  define signs  $s_i(u) = (-1)^{u \cdot h_i}$  and

$$x(u) = \frac{1}{2}(s_1(u), \dots, s_8(u)).$$

Select the 96-subset

$$S = \{u : u_7 = 0\} \cup \{u : u_7 = 1, u_1 \oplus u_2 \oplus u_3 = 0\}, \quad V = \{x(u) : u \in S\}.$$

Edges are declared by the  $E_8$  inner product:  $\langle x, y \rangle = 1$ .

### Published data

The 96-vector table and all invariants are provided as CSVs: `verified_E8_96_vectors.csv`, `verified_allpair_inner_products.csv`, `verified_edge_inner_products.csv`, `verified_nonedge_inner_products.csv`, `verified_degree_distribution.csv`, `verified_inner_product_counts.csv`, `verified_nonedge_inner_products.csv`.

### Verified invariants for the 96-set

All computations use exact rationals.

- $|V| = 96$ , and  $\langle x, x \rangle = 2$  for all  $x \in V$ .
- Distinct-pair inner-product multiset  $\left(\binom{96}{2}\right) = 4560$ :

$$-2 : 32, \quad -1 : 944, \quad 0 : 2544, \quad +1 : 1040.$$

- Graph statistics (edge iff  $\langle x, y \rangle = +1$ ):

$$\text{edges} = 1040, \text{ non-edges} = 3520, \text{deg}_{\min} = 16, \text{deg}_{\max} = 25, \overline{\text{deg}} = 65/3.$$

Antipodal pairs account for the 32 occurrences of  $-2$ ; this is consistent with  $-I \in W(E_8)$ .

### Normalized 64-set (no $-2$ among distinct pairs)

Fix a canonical representative per antipodal class by requiring the first nonzero coordinate to be  $+1/2$ ; then deduplicate. This yields  $V^{(64)}$  with  $|V^{(64)}| = 64$  and no antipodal collisions.

- Distinct-pair inner-product multiset  $\left(\binom{64}{2}\right) = 2016$ :

$$-1 : 224, \quad 0 : 1120, \quad +1 : 672.$$

- Graph is regular of degree 21; edges = 672, non-edges = 1344.

Exports: `normalized_64_vectors.csv`, `normalized_64_allpair_inner_products.csv`, `normalized_64_nonedge_inner_products.csv`, `normalized_64_degree_distribution.csv`, `normalized_64_inner_product_counts.csv`, `normalized_64_nonedge_inner_product_counts.csv`.

## Uniqueness up to $W(E_8)$

The map  $u \mapsto x(u)$  bijects  $\mathbb{F}_2^7$  with even-parity type-II sign patterns. The affine group  $G = \text{AGL}(7, 2) = 2^7 \rtimes \text{GL}(7, 2)$  acts by  $u \mapsto Au + t$ , inducing lattice automorphisms inside  $W(E_8)$  and preserving inner products. Our 96-set has the form

$$V = H_0 \cup \{u \in H_1 : m(u) = 0\}, \quad H_b = \{u : \ell(u) = b\},$$

with independent linear forms  $\ell(u) = u_7$  and  $m(u) = u_1 \oplus u_2 \oplus u_3$ . Since  $\text{GL}(7, 2)$  acts transitively on ordered independent pairs  $(\ell, m)$  and translations choose cosets, every subset of the form “one full hyperplane plus half of a disjoint hyperplane” is  $G$ -equivalent; hence all such 96-sets are  $W(E_8)$ -equivalent. Presence or absence of antipodals is inconsequential because  $-I \in W(E_8)$ .

## Reproducibility: exact-arithmetic test code

**Inputs.** A CSV `verified_E8_96_vectors.csv` with columns `x1, ..., x8` containing strings in  $\{“1/2”, “-1/2”\}$ .

**Python (exact checks and normalization).**

```
from fractions import Fraction
from itertools import combinations
import csv

# --- Load 96 vectors as Fractions ---
V = []
with open("verified_E8_96_vectors.csv", newline="") as f:
    rdr = csv.DictReader(f)
    cols = [f"x{i}" for i in range(1,9)]
    for row in rdr:
        v = [Fraction(1,2) if row[c].strip() == "1/2" else Fraction(-1,2) for c in cols]
        V.append(v)
assert len(V) == 96

# Convert to sign matrix S in {±1}
S = [[1 if a == Fraction(1,2) else -1 for a in v] for v in V]

# --- Exact checks for the 96-set ---
def ip(u, v): # exact inner product in E8
    s = sum(uu*vv for uu, vv in zip(u, v))    # integer in {-8,-6,...,8}
```

```

    return Fraction(s, 4)

# Lengths
assert all(ip(s, s) == 2 for s in S)

# Pairwise histogram
from collections import Counter
hist96 = Counter(ip(S[i], S[j]) for i, j in combinations(range(96), 2))
# Expected: {-2:32, -1:944, 0:2544, +1:1040}
print("96-set IP histogram:", {int(k): v for k, v in hist96.items()})

# Degrees for edges defined by <x,y>=1
deg96 = [0]*96
for i, j in combinations(range(96), 2):
    if ip(S[i], S[j]) == 1:
        deg96[i] += 1; deg96[j] += 1
print("deg min/max/avg:", min(deg96), max(deg96), sum(deg96)/96)

# --- Normalize to 64-set: fix first coord positive, then deduplicate ---
def canon(row):
    return tuple(row if row[0] == 1 else (-x for x in row))

seen = set()
S64 = []
for r in S:
    c = canon(r)
    if c not in seen:
        seen.add(c)
        S64.append(list(c))
assert len(S64) == 64

# Verify 64-set invariants
hist64 = Counter(ip(S64[i], S64[j]) for i, j in combinations(range(64), 2))
assert Fraction(-2, 1) not in hist64 # no antipodal pairs
print("64-set IP histogram:", {int(k): v for k, v in hist64.items()})

deg64 = [0]*64
for i, j in combinations(range(64), 2):
    if ip(S64[i], S64[j]) == 1:
        deg64[i] += 1; deg64[j] += 1
print("64-set degree uniform:", min(deg64), max(deg64), sum(deg64)/64)

```

Expected output (96-set).

$-2:32, \quad -1:944, \quad 0:2544, \quad +1:1040; \quad \deg_{\min} = 16, \deg_{\max} = 25, \overline{\deg} = 65/3.$

Expected output (64-set).

$$-1 : 224, \quad 0 : 1120, \quad +1 : 672; \quad \text{regular of degree 21.}$$

## 5 Restricted Initiality and Term Models for ResGraph

### 5.1 Signature, Axioms, and Models

Let  $\Sigma$  be the indexed total signature:

- A binary product symbol  $\otimes$  (disjoint orthogonal sum), with equations for commutativity, associativity, and unit; totality witnessed by indexing over orthogonal pairs.
- For each Dynkin automorphism  $\Gamma$  of a simply-laced simple system, a unary symbol  $\text{fold}_\Gamma$ .
- For each  $I \subseteq \{1, \dots, 8\}$ , a unary symbol  $\text{Par}_I$  (parabolic restriction).
- For each admissible extension index  $j$ , a unary symbol  $\text{Aug}_j$  (inverse parabolic augmentation).

Add 96 constants  $(a_k)_{k \in A}$  pinning a marking  $i : A \rightarrow X$ . Include finite-limit axioms  $\mathbf{L}$  enforcing:

1. The equations characterizing  $\otimes$ ,  $\text{fold}_\Gamma$ ,  $\text{Par}_I$ ,  $\text{Aug}_j$  and their compatibilities.
2. The reflection-closure of  $i(A)$  is a crystallographic root system of type  $E_8$  with Cartan relations of  $E_8$  (long-root norm 2).
3. Weyl-equivariance of these operations on the  $E_8$ -closure.

A  $\Sigma$ -model with marking (object of  $\mathbf{Alg}_{96}$ ) is a model of the finite-limit sketch determined by  $\Sigma$  and  $\mathbf{L}$  plus the constants  $(a_k)$ .

**Proposition 25** (Basic properties).  *$\mathbf{Alg}_{96}$  is locally presentable and monadic over  $\mathbf{Set}$ . Hence there is a finitary monad  $T$  on  $\mathbf{Set}$  with  $\mathbf{Alg}_{96} \simeq \mathbf{Set}^T$ , and free (term) models exist.*

### 5.2 Free/Term Model

Let  $F(A)$  be the term model on the 96-set  $A$ :

$$F(A) = \mathbf{Tm}_\Sigma(A) / \equiv_{\mathbf{L}}, \quad \iota : A \rightarrow F(A),$$

where  $\equiv_{\mathbf{L}}$  is the least congruence generated by  $\mathbf{L}$ .

**Theorem 26** (Restricted Initiality, pointed form). *For every  $\mathcal{A} \in \mathbf{Alg}_{96}$  interpreting  $(a_k)$  as a marking  $i : A \xrightarrow{\cong} S \subseteq \mathcal{A}$  whose reflection-closure is  $E_8$ , there exists a unique  $\Sigma$ -homomorphism*

$$\eta_{\mathcal{A}} : F(A) \rightarrow \mathcal{A}$$

*such that  $\eta_{\mathcal{A}} \circ \iota = i$ . The maps  $\eta_{\mathcal{A}}$  are natural in  $\mathcal{A}$ . Hence  $F(A)$  is initial in  $\mathbf{Alg}_{96}$ .*

*Proof sketch.* Define  $\eta_{\mathcal{A}}$  by structural recursion on  $\Sigma$ -terms and quotient by  $\equiv_{\mathbf{L}}$ . Uniqueness by induction on term depth. Naturality from functoriality of term interpretation.  $\square$

### 5.3 Unrestricted Initiality: Obstruction and Remedies

Let  $\text{Conf}_{96}(\mathcal{A})$  be the set of Atlas-type 96-subsets of  $\mathcal{A}$  whose reflection-closures are  $E_8$ . The Weyl group  $W(E_8)$  acts; typically  $\text{Conf}_{96}(\mathcal{A})$  is a (possibly empty)  $W(E_8)$ -torsor. A functorial map  $F(A) \rightarrow \mathcal{A}$  without a chosen marking exists iff this torsor has a natural global section, which fails when automorphisms permute distinct configurations.

**Conjecture 27** (Unrestricted Initiality). *For every  $\Sigma$ -model  $\mathcal{A}$  of  $\mathbf{L}$  there is a morphism  $F(A) \rightarrow \mathcal{A}$  unique up to automorphisms. Equivalently, every model carries a functorially natural Atlas-type 96-configuration.*

*Dependencies:* (C1) existence and Weyl-uniqueness of a 96-configuration in every simply-laced rank-8 subsystem; (C2) preservation/detection by the indexed operations.

**Forcing canonicity.** (i) Add an orientation/height to select one configuration; (ii) move to groupoid semantics and claim initiality up to isomorphism; (iii) add a functorial selector  $\sigma : \mathbf{1} \Rightarrow \text{Conf}_{96}$ .

### 5.4 Implementation Checklist

1. Enumerate index sets for  $\text{fold}_\Gamma$ ,  $\text{Par}_I$ ,  $\text{Aug}_j$  (totality).
2. List the finite equations forcing the  $E_8$  Cartan relations on the closure of  $i(A)$ .
3. Prove  $W(E_8)$ -equivariance of  $\text{fold}_\Gamma$ ,  $\text{Aug}_j$ ,  $\text{Par}_I$  on the closure.
4. Give a counterexample to unrestricted initiality: two disjoint 96-configurations swapped by an automorphism.

## 6 Group Action, Orbit Decomposition, and Invariant Functionals

### 6.1 Ambient Set and CRT Maps

Let  $P = \mathbb{Z}/48\mathbb{Z}$ ,  $B = \mathbb{Z}/256\mathbb{Z}$ , and  $G = P \times B$  so  $|G| = 48 \cdot 256 = 12,288$ . Write the Chinese Remainder isomorphism for  $P$  as

$$\psi : P \rightarrow \mathbb{Z}/16\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}, \quad \psi(p) = (a, c) = (p \bmod 16, p \bmod 3),$$

with inverse (encoding)

$$\varphi(a, c) \equiv 33a + 16c \pmod{48},$$

using  $3^{-1} \equiv 11 \pmod{16}$  and  $16^{-1} \equiv 1 \pmod{3}$ .

Represent  $a \in \mathbb{Z}/16\mathbb{Z}$  in binary as  $a = a_0 + 2a_1 + 4a_2 + 8a_3$  with  $a_i \in \{0, 1\}$ .

## 6.2 Reference Subgroup and Action

Let  $V \cong (\mathbb{Z}/2\mathbb{Z})^3$  act by flipping bits  $a_0, a_1, a_2$  (bit  $a_3$  fixed). Let  $W \cong (\mathbb{Z}/2\mathbb{Z})^8$  act on bytes by XOR. Define

$$U_{\text{ref}} = V \times W \cong (\mathbb{Z}/2\mathbb{Z})^{11}.$$

For  $u = (v, w) \in U_{\text{ref}}$  and  $g = (p, b) \in G$ :

1.  $(a, c) = \psi(p)$ .
2.  $a' = (a \oplus v)$  on bits  $a_0, a_1, a_2$  with  $a_3$  unchanged.
3.  $b' = b \oplus w$ .
4.  $p' = \varphi(a', c)$  and  $u \cdot g = (p', b')$ .

The 11 standard generators are commuting, each of order 2, giving a simple undirected Cayley graph on  $G$ .

## 6.3 Orbit Invariants, Freeness, and Tiling

Define the orbit invariant

$$\mathbf{o}(g) = (c, a_3) \in \mathbb{Z}/3\mathbb{Z} \times \{0, 1\}, \quad c = p \bmod 3, \quad a_3 = \lfloor a/8 \rfloor.$$

**Proposition 28.** *The action of  $U_{\text{ref}}$  on  $G$  is free. The invariant  $\mathbf{o}$  is preserved by  $U_{\text{ref}}$ . Each orbit has size  $2^{11} = 2048$  and is determined by  $(c, a_3)$ . There are exactly six orbits, with representatives*

$$S = \{(\varphi(0, c), 0), (\varphi(8, c), 0) : c \in \{0, 1, 2\}\}.$$

*Moreover, edges never cross orbits; each connected component of the Cayley graph equals a single orbit.*

*Sketch.* Nonzero  $v$  changes  $a$ ; nonzero  $w$  changes  $b$ ; hence stabilizers are trivial. The bits  $a_0, a_1, a_2$  and the byte are toggled freely, while  $c$  and  $a_3$  are fixed, so orbits have size  $2^{3+8} = 2048$  and are indexed by  $(c, a_3) \in \{0, 1, 2\} \times \{0, 1\}$ . The six anchors listed are in distinct orbits and tile  $G$ .  $\square$

## 6.4 Cayley Graph and Flag Complex

Let  $\Gamma$  be the Cayley graph of  $G$  with respect to the 11 generators. On each orbit,  $\Gamma$  is the 11-dimensional hypercube  $Q^{11}$ : per orbit  $n = 2^{11} = 2048$  vertices and  $m = 11 \cdot 2^{10} = 11264$  edges. The entire graph has six connected components, totaling

$$|V| = 12,288, \quad |E| = 67,584.$$

Hypercubes are triangle-free, thus the flag complex equals the 1-skeleton. For a graph,  $H_1$  is torsion-free and

$$\beta_1 = m - n + \#\text{components}.$$

Hence per orbit  $\beta_1 = 11264 - 2048 + 1 = 9217$ , and in total  $\beta_1 = 6 \cdot 9217 = 55,302$ . All higher Betti numbers vanish.

## 6.5 Schedule, Base Functional, and Invariance Scope

Let  $N = 768$  and the canonical fair schedule

$$\sigma(t) = (t \bmod 48, t \bmod 256), \quad t = 0, \dots, N-1.$$

For an affine observable  $f(p, b) = d + \alpha_p + \gamma_b$ , define the base functional

$$S_0(f) = \sum_{t=0}^{N-1} f(\sigma(t)) = 768 d + 16 \sum_{p \in P} \alpha_p + 3 \sum_{b \in B} \gamma_b.$$

**Proposition 29.**  *$S_0$  is invariant under arbitrary permutations of pages and bytes and under the  $U_{\text{ref}}$  action.*

[Limits for  $k \geq 1$ ] No polynomial weighting in  $t$  alone equalizes arithmetic-progression sums induced by  $\sigma$ . In particular, for  $\tau = t - (N-1)/2$ , the partial sums  $\sum_{t \equiv r (48)} \tau$  and  $\sum_{t \equiv s (256)} \tau$  are residue-dependent, so  $k \geq 1$  moment functionals are not invariant in general.

## 6.6 Permutation-Invariant Alternatives for First Moments

Define integer-scaled per-residue weights

$$w_p(t) = 2 \left\lfloor \frac{t}{48} \right\rfloor - 15, \quad w_b(t) = 2 \left\lfloor \frac{t}{256} \right\rfloor - 2,$$

so that for every page residue  $r$  and byte residue  $s$ ,

$$\sum_{i=0}^{15} w_p(r + 48i) = 0, \quad \sum_{i=0}^2 w_b(s + 256i) = 0.$$

Then the linear functionals

$$S_1^{\text{page}}(f) = \frac{1}{2} \sum_{t=0}^{N-1} w_p(t) f(\sigma(t)), \quad S_1^{\text{byte}}(f) = \frac{1}{2} \sum_{t=0}^{N-1} w_b(t) f(\sigma(t))$$

are invariant under arbitrary page permutations and byte permutations, respectively, and under the corresponding  $U_{\text{ref}}$  actions on  $P$  and  $B$ .

## 6.7 R96 Labeling

Define  $\lambda : B \rightarrow \{0, \dots, 95\}$  by  $\lambda(b) = b \bmod 96$ . Multiplicities are  $|\lambda^{-1}(r)| = 3$  for  $r \in \{0, \dots, 63\}$  and  $|\lambda^{-1}(r)| = 2$  for  $r \in \{64, \dots, 95\}$ , so  $64 \cdot 3 + 32 \cdot 2 = 256$ . Extend to  $G$  by composition with the byte projection; no equivariance beyond Weyl-invariance is asserted.

## 6.8 Operational Notes

Record the orbit tag  $\mathbf{o}(g)$  for indexing and caching. Since  $\mathbf{o}$  is invariant, edges never cross orbits and each component can be processed independently. Sparse  $\mathbb{F}_2$  or  $\mathbb{Z}$  homology on the 67,584-edge graph is straightforward; for graphs  $\beta_1$  is identical over  $\mathbb{Z}$  and  $\mathbb{F}_2$ .



## 6.9 E8-Based 96-Vertex Resonance Graph

Let  $R \subset \mathbb{R}^8$  be the  $E_8$  root system normalized by  $\langle \alpha, \alpha \rangle = 2$  for all  $\alpha \in R$ , so pairwise inner products lie in  $\{-2, -1, 0, 1, 2\}$ . Fix an injective indexing map

$$\iota : \{0, \dots, 95\} \longrightarrow R, \quad R_{96} := \text{im}(\iota),$$

and define the induced simple graph  $\Gamma_{E_8}^{(96)}$  by

$$V(\Gamma_{E_8}^{(96)}) = R_{96}, \quad \{\alpha, \beta\} \in E(\Gamma_{E_8}^{(96)}) \iff (\alpha \neq \beta \text{ and } \langle \alpha, \beta \rangle = +1).$$

No edges are added when  $\langle \alpha, \beta \rangle \in \{0, -1, 2, -2\}$ . This is the subgraph induced from the standard  $E_8$  root graph under the  $+1$  adjacency rule.

**Adjacency conventions.** The full  $E_8$  ResGraph elsewhere uses  $-1$  adjacency (edges for  $\langle \alpha, \beta \rangle = -1$ ). The 96-vertex graph here intentionally uses  $+1$  adjacency. Both conventions are Weyl-equivariant. Note also that in this subsection all roots satisfy  $\langle \alpha, \alpha \rangle = 2$  by normalization.

**Integration with the  $R_{96}$  labeling.** Use the existing byte partition to index  $R_{96}$  via

$$\lambda_{E_8} : B \rightarrow R_{96}, \quad \lambda_{E_8}(b) := \iota(b \bmod 96).$$

Then  $|\lambda_{E_8}^{-1}(\alpha)| = 3$  for  $\alpha \in \iota(\{0, \dots, 63\})$  and  $|\lambda_{E_8}^{-1}(\alpha)| = 2$  for  $\alpha \in \iota(\{64, \dots, 95\})$ , matching the multiplicities  $64 \cdot 3 + 32 \cdot 2 = 256$ . No compatibility with the  $U_{\text{ref}}$  action is required or claimed.

**Independence from the Cayley layer.**  $\Gamma_{E_8}^{(96)}$  lives solely on the label set and does not modify the Cayley graph  $\Gamma$  or its flag complex. It is an auxiliary resonance structure replacing any “96-dimensional” phrasing with the precise term “96-vertex subgraph of the  $E_8$  root graph (adjacency  $\langle \alpha, \beta \rangle = +1$ )”.

**Optional normalization.** One may choose  $R_{96}$  closed under negation ( $\alpha \in R_{96} \Rightarrow -\alpha \in R_{96}$ ); this adds no edges because  $\langle \alpha, -\alpha \rangle = -2$ , but can be convenient for symmetry book-keeping. No further properties of  $R_{96}$  are assumed.

## 7 Proofs (mathematical core)

**T1.** Product, quotient (fold), filtration, and augmentation in RESGRAPH output crystallographic root systems.

**T2.** For each output: existence of a simple system; Cartan integrality; Dynkin identification; *Weyl order = product of Coxeter exponents + 1*.

**T3.** Restricted initiality: the free RESGRAPH-algebra on the Atlas- $E_8$  96-seed is initial in the subcategory with a chosen 96-seed and laws **L**.

Conjectural: unrestricted initiality (clearly labeled as conjecture elsewhere).

## 8 Computational evidence (tests, not proofs)

- E1.** Atlas  $\rightarrow E_8$  96-set inner products over  $\mathbb{Q}$ : lengths 2; pair multiset on *unordered pairs with*  $\alpha \neq \beta$  is  $\{-2:32, -1:944, 0:2544, +1:1040\}$ . Identity check:  $32 + 944 + 2544 + 1040 = 4560 = \binom{96}{2}$ .
- E2.** Edge graph (adjacency  $\langle \alpha, \beta \rangle = +1$ ):  $|E| = 1040$ ; non-edges  $\binom{96}{2} - 1040 = 3520$ ; degree statistics min 16, max 25, average  $\bar{d} = 2|E|/|V| = 65/3$ .
- E3.** RESGRAPH operations executed symbolically with rationals: extracted simple systems; Cartans integral with off-diagonal products in  $\{0, 1, 2, 3\}$ ; Dynkin types match the canonical list; Weyl orders match products of exponents +1.

Scope: E1–E3 are numerical certificates; they guide but do not replace proofs.

## 9 Artifact manifest

### Data (inputs, canonical CSVs)

- `verified_E8_96_vectors.csv` — 96 half-integer roots, fixed order.
- `iota_96.csv` — explicit indexing map  $\iota$  (ordering of the 96 roots).
- `verified_allpair_inner_products.csv` — full 4560 counts over  $\{-2, -1, 0, +1\}$ .
- `verified_edge_inner_products.csv` — pairs with inner product +1.
- `verified_nonedge_inner_products.csv` — pairs with inner product in  $\{-2, -1, 0\}$ .
- `verified_degree_distribution.csv` — vertex degrees for the edge graph.
- `verified_inner_product_counts.csv` — global multiplicities.
- `verified_nonedge_inner_product_counts.csv` — non-edge multiplicities.
- `adjacency_plus1_96.npz` — sparse CSR of  $\Gamma_{E_8}^{(96)}$  under +1 adjacency.

### Derived (deterministic from the above)

- `cartan*.json` — Cartan matrices per operation instance.
- `dynkin*.txt` — recognized Dynkin type and rank.
- `weyl_orders*.txt` — Weyl-group orders.
- `resgraph_ops_spec.md` — machine-readable op trees producing each output.
- `checks_report.md` — pass/fail per proof obligation.

### Reproducibility

Record SHA-256 for every file; fix rational arithmetic (no floats); fix ordering: lexicographic on vectors, then on pairs.

## 10 Formal-verification plan (Lean and Coq)

### Targets (proof obligations)

- P0.** Exact model of  $\mathbb{Q}^8$  with bilinear form and reflections.
- P1.** 96-set correctness: each row has even parity; norm 2.
- P2.** Pairwise inner products in  $\{-2, -1, 0, +1\}$  with counts as in E1.
- P3.** RESGRAPH operations preserve the “crystallographic root system” predicate.
- P4.** For every output: existence of a simple system by constructive algorithm; Cartan integral with  $a_{ii} = 2$  and  $a_{ij}a_{ji} \in \{0, 1, 2, 3\}$ .
- P5.** Dynkin type equals the canonical type of the computed Cartan (decision procedure).
- P6.** Weyl order equals product of Coxeter exponents +1 of that type.
- P7.** Restricted initiality: existence and uniqueness (mod Weyl) of the RESGRAPH-algebra morphism from the free algebra on the 96-seed.

### Lean (mathlib) roadmap

Define `root_vec` := `vector 8`, an inner product `ip`, reflections `refl x := x - 2*(ip x / ip ) •`, and predicates `is_rootset`, `is_crystallographic`, `simple_system X`, `cartan`. Avoid IO in proofs; re-derive CSV contents. Use `linarith`, `norm_num`, `ring`, finite enumerations via `finset`. Implement folds, filters, augments, Cartan→Dynkin decision tree, and Weyl-order tables with proofs. Prove initiality by structural induction on syntax trees modulo `L`.

### Coq (QArith + MathComp) roadmap

Mirror the Lean development using `QArith` rationals and `MathComp` matrices; use small-scale reflection for finite decisions.

### Tooling pins

Record exact versions: Lean (version and mathlib SHA), Coq (version, QArith/MathComp SHAs).

## 11 Separation: proof vs. evidence

Proof artifacts live in Lean/Coq files with completed theorems. Evidence artifacts live as CSV/MD plus reproducible generators. Reports label every claim as PROVED or EVIDENCE with pointers to files.

## A Machine check for $G_2$

*# G2 fold inside E8: minimal checks*

```
from fractions import Fraction
import math
```

```
# Gram on span{ beta_l = a1+a2+a4, beta_s = a3 }:
# /beta_l/^2=6, /beta_s/^2=2, (beta_l,beta_s)=-3
G = [[Fraction(6,1), Fraction(-3,1)],
      [Fraction(-3,1), Fraction(2,1)]]
```

```
def dot(c,d): # c,d in coefficient basis [beta_l, beta_s]
    return c[0]*(G[0][0]*d[0] + G[0][1]*d[1]) + c[1]*(G[1][0]*d[0] + G[1][1]*d[1])
```

```
def l2(c): return dot(c,c)
```

*# simple roots in this rank-2 subsystem*

```
aL = [Fraction(1,1), Fraction(0,1)] # beta_l
aS = [Fraction(0,1), Fraction(1,1)] # beta_s
```

*# Cartan: A\_ij = 2\*(ai,aj)/(aj,aj)*

```
def A(i,j):
    ai = aL if i==1 else aS
    aj = aL if j==1 else aS
    return 2*dot(ai,aj)/l2(aj)
```

```
A11, A12 = A(1,1), A(1,2)
A21, A22 = A(2,1), A(2,2)
print("Cartan:", [[int(A11), int(A12)], [int(A21), int(A22)]])
```

```
# angle and Coxeter m: theta = arccos( (aL,aS)/(/aL//aS/) )
cos_th = float(dot(aL,aS) / math.sqrt(float(l2(aL)*l2(aS))))
theta = math.acos(cos_th) # in radians
m = round(math.pi / (math.pi - theta)) # since theta = pi - pi/m
print("cos(theta)=", f"{cos_th:.12f}")
print("theta_deg=", theta*180.0/math.pi)
print("m=", m)
```

*# reflections and orbit*

```
def refl(v, i): # reflect v across ai
    ai = aL if i==1 else aS
    return [v[0] - 2*dot(v,ai)/l2(ai)*ai[0],
            v[1] - 2*dot(v,ai)/l2(ai)*ai[1]]
```

```
roots = set()
front = {tuple(aL), tuple(aS), tuple([-aL[0], -aL[1]]), tuple([-aS[0], -aS[1]])}
```

```

while front:
    new = set()
    for v in front:
        if v not in roots:
            roots.add(v)
            new.add(tuple(refl(list(v),1)))
            new.add(tuple(refl(list(v),2)))
    front = new

# count lengths
from collections import Counter
len_counts = Counter(l2(list(v)) for v in roots)
print("root_count_=", len(roots))
print("length^2_ classes:", {float(k):v for k,v in len_counts.items()})

# Weyl group size via matrices
def refl_matrix(i):
    I = [[Fraction(1,1),Fraction(0,1)],[Fraction(0,1),Fraction(1,1)]]
    row = G[i-1]
    if i==1: e = [Fraction(1,1), Fraction(0,1)]
    else:     e = [Fraction(0,1), Fraction(1,1)]
    M = [[(Fraction(2,1)/G[i-1][i-1]) * e[r] * row[c] for c in range(2)] for r in range(2)]
    return [[I[r][c] - M[r][c] for c in range(2)] for r in range(2)]

def matmul(A,B):
    return [[A[0][0]*B[0][0] + A[0][1]*B[1][0], A[0][0]*B[0][1] + A[0][1]*B[1][1],
            A[1][0]*B[0][0] + A[1][1]*B[1][0], A[1][0]*B[0][1] + A[1][1]*B[1][1]]

def mateq(A,B):
    return all(A[r][c]==B[r][c] for r in range(2) for c in range(2))

S1, S2 = refl_matrix(1), refl_matrix(2)
group = [[[Fraction(1,1),Fraction(0,1)],[Fraction(0,1),Fraction(1,1)]]]
queue = [S1, S2]
while queue:
    M = queue.pop()
    if not any(mateq(M,N) for N in group):
        group.append(M)
        queue += [matmul(M,S1), matmul(M,S2), matmul(S1,M), matmul(S2,M)]
print("Weyl_order_=", len(group))

# Expected:
# Cartan: [[2, -3], [-1, 2]]
# m = 6
# root_count = 12 with length^2 classes {2:6, 6:6}
# Weyl_order = 12

```

## B Machine check for $F_4$

```
# F4 via E6 quotient fold inside E8: minimal checks

from fractions import Fraction
import math
from collections import Counter

# Gram on {beta1,beta2,beta3,beta4}:
# /b1/^2=/b2/^2=1, /b3/^2=/b4/^2=2
# (b1,b2)=-1/2, (b2,b3)=-1, (b3,b4)=-1, others 0
G = [[Fraction(1,1), Fraction(-1,2), Fraction(0,1), Fraction(0,1)],
      [Fraction(-1,2), Fraction(1,1), Fraction(-1,1), Fraction(0,1)],
      [Fraction(0,1), Fraction(-1,1), Fraction(2,1), Fraction(-1,1)],
      [Fraction(0,1), Fraction(0,1), Fraction(-1,1), Fraction(2,1)]]

n = 4
def dot(c,d): return sum(c[i]*G[i][j]*d[j] for i in range(n) for j in range(n))
def l2(c): return dot(c,c)

# simple roots in this folded basis
e = [[Fraction(1,1) if i==k else Fraction(0,1) for k in range(n)] for i in range(n)]

# Cartan A_ij = 2*(alpha_i,alpha_j)/(alpha_j,alpha_j)
A = [[2*dot(e[i],e[j])/l2(e[j]) for j in range(n)] for i in range(n)]
print("Cartan:")
for r in A: print([int(x) for x in r])

# Coxeter m_ij from product a_ij a_ji in {0,1,2,3}
def m_from_prod(p): return {0:2, 1:3, 2:4, 3:6}[p]
for (i,j) in [(0,1),(1,2),(2,3)]:
    p = A[i][j]*A[j][i]
    print(f"m_{{{i+1}}}{j+1}}_=" , m_from_prod(int(p)))

# reflections and orbit
def refl(v,i): return [v[k] - 2*dot(v,e[i])/l2(e[i])*e[i][k] for k in range(n)]
roots, front = set(), set()
for i in range(n):
    front.add(tuple(e[i])); front.add(tuple([-x for x in e[i]]))
while front:
    new=set()
    for v in front:
        if v not in roots:
            roots.add(v)
            for i in range(n): new.add(tuple(refl(list(v),i)))
    front=new
```

```

cnt = Counter(l2(list(v)) for v in roots)
print("root_count_=", len(roots))
print("length^2_classes:", {float(k):v for k,v in cnt.items()})

# Weyl group order from four reflections
def reflect_matrix(i):
    I = [[Fraction(1,1) if r==c else Fraction(0,1) for c in range(n)] for r in range(n)]
    S = [row[:] for row in I]
    row = G[i]
    for r in range(n):
        for c in range(n):
            S[r][c] -= (Fraction(2,1)/G[i][i])*(Fraction(1,1) if r==i else Fraction(0,1))
    return S

def matmul(A,B):
    C=[[Fraction(0,1) for _ in range(n)] for _ in range(n)]
    for i in range(n):
        for k in range(n):
            for j in range(n):
                C[i][j]+=A[i][k]*B[k][j]
    return C

def mateq(A,B): return all(A[r][c]==B[r][c] for r in range(n) for c in range(n))

S=[reflect_matrix(i) for i in range(n)]
group=[[Fraction(1,1) if r==c else Fraction(0,1) for c in range(n)] for r in range(n)]
queue=S[:]
while queue:
    M=queue.pop()
    if not any(mateq(M,N) for N in group):
        group.append(M)
        for X in S:
            queue += [matmul(M,X), matmul(X,M)]
print("Weyl_order_=", len(group))

# Expected:
# Cartan =
# [2, -1, 0, 0]; [-1, 2, -1, 0]; [0, -2, 2, -1]; [0, 0, -1, 2]
# m_12=3, m_23=4, m_34=3
# root_count=48 with length^2 {1:24, 2:24}
# Weyl_order=1152

```

## C Machine check $E_6$ (filtration)

```
# E6 via standard parabolic inside E8: minimal checks
```

```

from fractions import Fraction
from collections import Counter

# Bourbaki E6: trivalent at 4, edges:
# (1-3), (3-4), (4-5), (5-6), and arm (2-4)
n = 6
edges = {(1,3),(3,4),(4,5),(5,6),(2,4)}
edges |= {(j,i) for (i,j) in edges}

# Gram matrix: 2 on diag, -1 on edges, 0 else
G = [[Fraction(2,1) if i==j else
      (Fraction(-1,1) if (i+1,j+1) in edges else Fraction(0,1))
      for j in range(n)] for i in range(n)]

def dot(c,d): return sum(c[i]*G[i][j]*d[j] for i in range(n) for j in range(n))
def l2(c): return dot(c,c)

# simple roots
e = [[Fraction(1,1) if i==k else Fraction(0,1) for k in range(n)] for i in range(n)]

# Cartan matrix
A = [[2*dot(e[i],e[j])/l2(e[j]) for j in range(n)] for i in range(n)]
print("Cartan matrix A:")
for r in A: print([int(x) for x in r])

# det(A) and rank over Q
def det_rational(M):
    from copy import deepcopy
    M = [[Fraction(m,1) for m in row] for row in M]
    det = Fraction(1,1); m = len(M)
    for i in range(m):
        p = next((r for r in range(i,m) if M[r][i]!=0), None)
        if p is None: return Fraction(0,1)
        if p!=i: M[i],M[p]=M[p],M[i]; det*=-1
        det *= M[i][i]
        piv = M[i][i]
        for c in range(i,m): M[i][c] /= piv
        for r in range(i+1,m):
            f = M[r][i]
            if f!=0:
                for c in range(i,m): M[r][c] -= f*M[i][c]
    return det

def rank_rational(M):
    from copy import deepcopy
    M = [[Fraction(m,1) for m in row] for row in M]
    r=c=0; R=len(M); C=len(M[0])

```



```

while r<R and c<C:
    p = next((i for i in range(r,R) if M[i][c]!=0), None)
    if p is None: c+=1; continue
    M[r],M[p]=M[p],M[r]; piv=M[r][c]
    for j in range(c,C): M[r][j]/=piv
    for i in range(R):
        if i!=r and M[i][c]!=0:
            f=M[i][c]
            for j in range(c,C): M[i][j]-=f*M[r][j]
    r+=1; c+=1
return r

print("det(A)□=", det_rational(A), "rank(A)□=", rank_rational(A))

# Reflection closure
def refl(v,i):
    ai=e[i]
    return [v[k]-2*dot(v,ai)/l2(ai)*ai[k] for k in range(n)]

roots=set(); front=set()
for i in range(n):
    front.add(tuple(e[i])); front.add(tuple([-x for x in e[i]]))
while front:
    new=set()
    for v in front:
        if v not in roots:
            roots.add(v)
            for i in range(n): new.add(tuple(refl(list(v),i)))
    front=new

cnt = Counter(l2(list(v)) for v in roots)
print("root_count□=", len(roots))
print("length2□classes:", {float(k):v for k,v in cnt.items()})

# Weyl order from exponents
exponents = [1,4,5,7,8,11]
order = 1
for m in exponents: order *= (m+1)
print("|W(E6)|□=", order)

# Expected:
# det(A)=3, rank=6
# root_count=72, all length2=2
# |W(E6)|=51840

```

## D Machine check $E_7$ (augmentation)

```

# E7 via standard parabolic inside E8 with augmentation at the junction

from fractions import Fraction
from collections import Counter

# Bourbaki E7 inside E8: trivalent at node 4
# edges: (1-3), (3-4), (4-5), (5-6), (6-7) and arm (2-4)
n = 7
edges = {(1,3),(3,4),(4,5),(5,6),(6,7),(2,4)}
edges |= {(j,i) for (i,j) in edges}

# Gram matrix: 2 on diag, -1 on edges, 0 else
G = [[Fraction(2,1) if i==j else
      (Fraction(-1,1) if (i+1,j+1) in edges else Fraction(0,1))
      for j in range(n)] for i in range(n)]

def dot(c,d): return sum(c[i]*G[i][j]*d[j] for i in range(n) for j in range(n))
def l2(c): return dot(c,c)

# simple roots basis
e = [[Fraction(1,1) if i==k else Fraction(0,1) for k in range(n)] for i in range(n)]

# Cartan matrix
A = [[2*dot(e[i],e[j])/l2(e[j]) for j in range(n)] for i in range(n)]
print("Cartan matrix A:")
for r in A: print([int(x) for x in r])

# det(A) and rank over Q
def det_rational(M):
    M = [[Fraction(m,1) for m in row] for row in M]
    det = Fraction(1,1); m = len(M)
    for i in range(m):
        p = next((r for r in range(i,m) if M[r][i]!=0), None)
        if p is None: return Fraction(0,1)
        if p!=i: M[i],M[p]=M[p],M[i]; det*=-1
        piv = M[i][i]; det *= piv
        for c in range(i,m): M[i][c] /= piv
        for r in range(i+1,m):
            f = M[r][i]
            if f!=0:
                for c in range(i,m): M[r][c] -= f*M[i][c]
    return det

def rank_rational(M):
    M = [[Fraction(m,1) for m in row] for row in M]

```

```

r=c=0; R=len(M); C=len(M[0])
while r<R and c<C:
    p = next((i for i in range(r,R) if M[i][c]!=0), None)
    if p is None: c+=1; continue
    M[r],M[p]=M[p],M[r]; piv=M[r][c]
    for j in range(c,C): M[r][j]/=piv
    for i in range(R):
        if i!=r and M[i][c]!=0:
            f=M[i][c]
            for j in range(c,C): M[i][j]-=f*M[r][j]
    r+=1; c+=1
return r

print("det(A)□=", det_rational(A), "rank(A)□=", rank_rational(A))

# Reflection-closure
def refl(v,i):
    ai=e[i]
    return [v[k]-2*dot(v,ai)/l2(ai)*ai[k] for k in range(n)]

roots,setfront=set(),set()
front=set()
for i in range(n):
    front.add(tuple(e[i]))
    front.add(tuple([-x for x in e[i]]))

while front:
    new=set()
    for v in front:
        if v not in roots:
            roots.add(v)
            for i in range(n):
                new.add(tuple(refl(list(v),i)))
    front=new

cnt = Counter(l2(list(v)) for v in roots)
print("root_count□=", len(roots))
print("length^2□classes:", {float(k): v for k,v in cnt.items()})

# Weyl order from exponents
exponents = [1,5,7,9,11,13,17]
order = 1
for m in exponents: order *= (m+1)
print("|W(E7)|□=", order)

# Expected:
# det(A)=2, rank=7

```

```
# root_count=126, all length^2=2
# |W(E7)|=2,903,040
```

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